Stanley Nelavala

Software Engineer: AI Operations, Internal Tools, Agents and MCP

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Huntington, IN | Open to relocate anywhere

Summary

Full-stack Python and TypeScript engineer who builds internal platforms and agents so teams do more without more people. At a 10-person startup I designed the signed service-to-service API (auth) every module uses, automated AWS start/stop and malware-scan controls across the platform, and shipped web and mobile products. I build agents and MCP tools (LangGraph, evals, audit logs), and I use Claude Code, Codex and opencode every day with written acceptance checks. M.S. in Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Automated weekday start and stop of dev and acceptance servers and databases (ECS, RDS): about 53 hours a week instead of 168. Standardized malware scanning of file uploads (GuardDuty) across 9 app modules.
- Built RxDesign, the hospital design service, backend and web (FastAPI, React, TypeScript), with a deterministic, rule-based layout engine and an LLM setting; 220+ offline tests on the API.
- Built the RxAsset iOS and Android app (React Native, Expo) with scan-to-search and an on-device voice assistant; set up iOS and Android release pipelines in GitHub Actions with automatic TestFlight submission.
- Replaced offset paging with keyset paging on the catalog device list (285 ms to 0.20 ms at row 150,000 on a 328,000-device dev database) and cut the planning Items page from 3 API calls and 162 KB to 2 calls and 37 KB.
- Use Claude Code, Codex and opencode every day to design, write, debug and review code; tests and written acceptance checks gate every change.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session.

Projects (public code, tests)

Agent Integration Platform | *Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis, Terraform* GitHub

- LangGraph case workflow with human approval that resumes after restart, step and token budgets and one constrained MCP write tool; retrieval with access labels so an agent sees only what it may.
- 62 deterministic replay evals, prompt-injection tests, PII redaction, an append-only audit log, and failover between two LLM adapters with a circuit breaker and token cost records.

Customer Integration Console | *TypeScript, Next.js, Fastify, PostgreSQL, OIDC, Keycloak, Docker* GitHub

- Internal web console with single sign-on (OIDC, Keycloak realm config) in front of a Fastify and PostgreSQL API, image scanned with Trivy.

Agent-fleet build workflow | *Claude Code, opencode, DeepSeek, shell* GitHub

- A planner, manager and worker agent pipeline that delivered two public services and an ops console in 34 work packets, each with written acceptance checks and a progress log.

Skills

Agents and AI ops: LangGraph, MCP tools, deterministic evals, guardrails, PII redaction, LLM failover, internal tooling and workflow automation; Claude Code, Codex, opencode, a DeepSeek harness
Languages: Python, TypeScript, JavaScript, SQL, Java, Go, Rust
Web and APIs: FastAPI, React, Next.js, Node.js, REST, OIDC and service-to-service auth, rate limiting, keyset paging
Data and cloud: PostgreSQL, Redis, AWS (ECS, RDS, S3, SQS, GuardDuty), Terraform, Docker, GitHub Actions
Quality: pytest, Playwright, k6, OpenTelemetry, acceptance checks, review of agent output

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)